

Maciej Karczmarz

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📍 Stalowa Wola, Poland

🐙 [Github](#)

🌐 [LinkedIn](#)

🎨 [ArtStation](#)

Projects

Programming Projects

Gix (Go) – Developed a real-time currency exchange rate tool that aggregates data from local exchange offices for accurate/up-to-date currency tracking.

GRAT (JavaFX) – Interactive weather visualization system. The application integrates OpenWeatherMap, Google Maps and Windy APIs to deliver real-time environmental data alongside precise location context.

GRATS (C++, OpenGL) – Created a minimalist graphics renderer to provide a learning platform for OpenGL and graphics programming fundamentals.

Sand-Simulation (C++) – Granular physics engine focused on realistic particle behavior. The project simulates natural environmental interactions, emphasizing efficient computation and physical accuracy.

SmartOffer– Automation system for AB Bechicki. It transforms inquiries into ready-to-use cost estimates (Premium/Budget) by selecting materials based on technical standards and parameters, not just brands. Built on RAG architecture, n8n workflows, and AWS, ensuring enterprise-grade scalability.

Graphics Projects

Unreal Engine 5 Project – Developing realistic environments using Gaea 2.0 and Houdini tools, with focus on procedural workflows and software integration.

ArtStation Profile – Created a variety of 3D assets and visualizations, ranging from realistic architectural renders and product visualizations to stylized fantasy scenes.

Languages

Polish

Native

English

B2+

German

A1

About me

Backend & 3D Graphics Developer – I have reasonable experience in backend development (mainly Go) and in creating and optimizing 3D assets. My projects range from backend applications and automation tools to 3D modeling, texturing, and rendering, sometimes combining both areas (e.g., interactive or real-time 3D content). I aim to work as a backend developer (with a focus on Cloud/Fin Ops) or in roles that connect 3D graphics with programming.

Experience

Simplicity Games

Technical Artist

Rzeszów, Poland

November 2024 - January 2025

3D Model Creation & Optimization – designing models in Blender from scratch and based on reference materials, creating optimized low-poly meshes, LODs, and collision meshes. Composing and processing PBR textures using tools such as Substance Painter, baking normal maps. Optimizing assets for performance and compliance with Unity Engine requirements, including UV improvements.

<https://simplicitygames.pl/>

Education

Rzeszow University of Technology

Computer Science

October 2022 - February 2026

Bachelor's Studies – Faculty of Electrical and Computer Engineering, Rzeszow University of Technology

Engaged in a variety of academic and additional projects combining software development, data analysis, web technologies and 3D graphics. Key activities and achievements include:

- **C++ development** – designed and implemented a graphical simulation project as part of coursework.
- **Data Analysis** – applied Python, MATLAB, and R for data processing, visualization, and statistical modeling.
- **Team-based Web Development** – collaborated with peers from the faculty on a group project using React and modern web technologies.
- **AI Research Project** – developed an experimental AI solution (MLP Algorithm variants) within the Artificial Intelligence course, exploring machine learning concepts.
- **Hackathon Participation** – took part in events such as NASA Space Apps Challenge, DIGIEDUHACK, Hack SPACESHIELD or Hack CARPATHIA, focusing on rapid prototyping and problem-solving under time constraints.
- **Student Research Group** – member of SKNI KOD since 2022, contributing to multiple collaborative and independent projects in software engineering and 3D graphics.

Public High School, Diploma

STEM (Mathematics, Physics, Computer Science)

September 2019 - May 2022

- Extended Mathematics: 80% (Standard Mathematics: 100%)
- Extended English: 84% (Standard English: 98%)
- Extended Computer Science: 53%
- Extended Physics: 50%